The Discrete–Continuous Bridge Through the Lemniscatic Lens:

An Analysis of $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2}$

Abstract

We present a complete derivation of the closed-form identity

$$\sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2} \;=\; \frac{2 + \pi \coth \pi + \pi^2 \operatorname{csch}^2 \pi}{4},$$

then recast the result entirely in terms of the lemniscate constant ϖ and the bridge constant $G^* = 2\varpi/\sqrt{\pi}$ of Foundational Ternary Dynamics (FTD). In the FTD rewriting, the ontic content of the identity reduces to the single ratio $\ell = \varpi^2/{G^*}^2 = \pi/4$, revealing that every appearance of π in the standard formula is a derived consequence of this deeper lemniscatic ratio.

We show that the sum is a restriction of the Epstein zeta function $Z_{\mathbb{Z}[i]}(s)$ at $s = 2$, establish the exact identity $E_4(i) = 3\,{G^*}^8/(256\,\varpi^4)$, and prove the Epstein factorisation $Z_{\mathbb{Z}[i]}(2) = 4\,\zeta(2)\,L(2,\chi_{-4}) = 32\,\varpi^4\,G/(3\,{G^*}^4)$ in ontic variables, where G is Catalan's constant—the irreducible arithmetic residue encoding the distribution of Gaussian primes.

We identify the exponential scale $e^{4\ell} = e^{\pi}$ (Gelfond's constant) as the natural decay parameter controlling the lattice's discrete-to-continuous transition, and show that the row restriction from $\mathbb{Z}[i]$ to a single horizontal line eliminates the arithmetic factor, explaining why our one-dimensional sum admits a purely ontic closed form.

We then place these results in a dimensional hierarchy: the Basel sum $\zeta(2) = \pi^2/6$ is the lattice zeta function of $\mathbb{Z}$ at $s = 2$; the Eisenstein series G_4 encodes the period of $\mathbb{Z}[i]$ through ϖ; the Watson integral encodes the period of $\mathbb{Z}^3$ through G^*. Each dimension has its own "circle constant," and only $D = 3$ produces a self-consistent framework.

This paper is one of five companion papers presenting Foundational Ternary Dynamics; see [2] for the derivation chain, [3] for the physical lifecycle, [4] for the ontological constructor, and [5] for the computational implementation.

Contents

1 The Problem

We seek a closed form for the sum

$$S \;=\; \sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2}\,. \tag{1}$$

This is a convergent series of rational terms indexed by the non-negative integers. Its evaluation requires passing from the discrete lattice $\mathbb{N}$ into the analytic structure of the complex plane, and the result is a transcendental number expressed in terms of hyperbolic functions of π. The purpose of this note is to make the architecture of that passage explicit and to identify its ontic content.

2 Standard Derivation

2.1 Strategy

The exponent 2 in the denominator of (1) makes direct summation difficult. We proceed by first evaluating the simpler sum $\sum 1/(n^2+a)$ in closed form, then recovering S by differentiation with respect to the auxiliary parameter a.

2.2 The Bilateral Sum

The Mittag-Leffler expansion of $\pi\cot(\pi z)$ gives the classical identity

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+a} \;=\; \frac{\pi}{\sqrt{a}}\coth(\pi\sqrt{a})\,, \qquad a>0. \tag{2}$$

This can be derived from the partial fraction decomposition of $\pi\cot(\pi z)$, or equivalently from the Poisson summation formula applied to the Lorentzian $f(x) = 1/(x^2+a)$.

Remark 2.1 (First bridge: lattice to analysis)**.** *Equation* (2) *is already a discrete-to-continuous bridge: the left side is a sum over the integer lattice $\mathbb{Z}$, while the right side is a smooth function of the continuous parameter a, expressed in terms of the hyperbolic cotangent. The Poisson summation route makes the mechanism explicit: the Fourier transform converts the sum over direct-lattice points into a sum over reciprocal-lattice points, and the latter is dominated by a single exponential scale.*

2.3 The Unilateral Sum

Since $1/(n^2+a)$ is symmetric in n, the bilateral sum double-counts every term except the $n = 0$ term. Hence

$$\sum_{n=0}^{\infty} \frac{1}{n^2+a} \;=\; \frac{1}{2}\left(\frac{\pi}{\sqrt{a}}\,\coth(\pi\sqrt{a}) + \frac{1}{a}\right) \;=\; \frac{\pi\coth(\pi\sqrt{a})}{2\sqrt{a}} + \frac{1}{2a}\,. \tag{3}$$

Note that every term on the left is positive, and the right side is manifestly positive for $a > 0$, as required.

2.4 Differentiation with Respect to the Parameter

Observe that

$$\frac{1}{(n^2+a)^2} \;=\; -\frac{\partial}{\partial a}\left(\frac{1}{n^2+a}\right). \tag{4}$$

Summing over $n \geq 0$ and exchanging the sum and derivative (justified by uniform convergence on compact subsets of $(0,\infty)$):

$$\sum_{n=0}^{\infty} \frac{1}{(n^2+a)^2} \;=\; -\frac{d}{da}\sum_{n=0}^{\infty} \frac{1}{n^2+a}\,. \tag{5}$$

Define

$$g(a) \;=\; \frac{\pi\coth(\pi\sqrt{a})}{2\sqrt{a}} + \frac{1}{2a}\,. \tag{6}$$

Then

$$S \;=\; -\,g'(1). \tag{7}$$

Remark 2.2 (Second bridge: discrete to continuous)**.** *The fixed problem at $a = 1$ has been embedded into a one-parameter continuous family. The answer is recovered by differentiation followed by evaluation, converting a discrete summation problem into a calculus problem.*

2.5 Computing $g'(a)$

Write $g = g_1 + g_2$ with

$$g_1(a) = \frac{\pi\coth(\pi\sqrt{a})}{2\sqrt{a}}\,, \qquad g_2(a) = \frac{1}{2a}\,.$$

For g_2:

$$g_2'(a) = -\frac{1}{2a^2}\,. \tag{8}$$

For g_1, apply the product rule to $\frac{\pi}{2}\cdot\coth(\pi\sqrt{a})\cdot a^{-1/2}$:

$$\begin{aligned} g_1'(a) &= \frac{\pi}{2}\left[\frac{d}{da}\big(\coth(\pi\sqrt{a})\big)\cdot a^{-1/2} + \coth(\pi\sqrt{a})\cdot\big(-\tfrac{1}{2}\big)a^{-3/2}\right] \\ &= \frac{\pi}{2}\left[\big(-\operatorname{csch}^2(\pi\sqrt{a})\big)\cdot\frac{\pi}{2\sqrt{a}}\cdot\frac{1}{\sqrt{a}} - \frac{\coth(\pi\sqrt{a})}{2a^{3/2}}\right] \\ &= -\frac{\pi^2\operatorname{csch}^2(\pi\sqrt{a})}{4a} - \frac{\pi\coth(\pi\sqrt{a})}{4a^{3/2}}\,. \end{aligned} \tag{9}$$

Combining:

$$g'(a) \;=\; -\frac{\pi^2\operatorname{csch}^2(\pi\sqrt{a})}{4a} - \frac{\pi\coth(\pi\sqrt{a})}{4a^{3/2}} - \frac{1}{2a^2}\,. \tag{10}$$

2.6 Evaluation at $a = 1$

Setting $a = 1$ in (10):

$$g'(1) = -\frac{\pi^2 \operatorname{csch}^2 \pi}{4} - \frac{\pi \coth \pi}{4} - \frac{1}{2} = -\frac{\pi^2 \operatorname{csch}^2 \pi + \pi \coth \pi + 2}{4}.$$

By (7):

$$\boxed{S \;=\; \sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2} \;=\; \frac{2 + \pi \coth \pi + \pi^2 \operatorname{csch}^2 \pi}{4} \;\approx\; 1.30683697542\,.} \tag{11}$$

3 FTD Recasting: Eliminating π

3.1 The Ontic Constants

Definition 3.1 (Lemniscate constant)**.** *The lemniscate constant ϖ is defined by*

$$\varpi \;=\; 2\int_0^1 \frac{dt}{\sqrt{1-t^4}} \;=\; \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}} \;\approx\; 2.62206\,.$$

It is the half-perimeter of the lemniscate of Bernoulli and satisfies $\varpi = \pi/M(1, \sqrt{2})$, where M denotes the arithmetic–geometric mean.

Definition 3.2 (Bridge constant)**.** *The FTD bridge constant is*

$$G^* \;=\; \frac{\Gamma(1/4)^2}{\sqrt{2}\,\Gamma(1/2)^2} \;=\; \frac{2\varpi}{\sqrt{\pi}} \;\approx\; 2.95868\,.$$

The first form is π-free: both $\Gamma(1/4)$ and $\Gamma(1/2)$ are definable by π-free integrals ($\int_0^\infty t^{s-1}e^{-t}\,dt$ at $s = 1/4$ and $s = 1/2$). The equivalence of the two forms follows from $\Gamma(1/2)^2 = \pi$ (the Gaussian integral).

Definition 3.3 (Ontic ratio)**.** *Define*

$$\ell \;=\; \frac{\varpi^2}{G^{*2}}\,. \tag{12}$$

Lemma 3.4. *$\pi = 4\ell = 4\varpi^2/G^{*2}$.*

Proof. From $G^* = 2\varpi/\sqrt{\pi}$ we obtain $G^{*2} = 4\varpi^2/\pi$, hence $\pi = 4\varpi^2/G^{*2} = 4\ell$. □

Remark 3.5. *In the FTD framework, ϖ and G^* (equivalently ϖ and ℓ) are treated as the ontic (structurally primary) constants. The number π is a derived quantity: it is four times the ratio ϖ^2/G^{*2}. This reflects the claim that the lemniscatic lattice structure is ontologically prior to the circular one.*

3.2 The Sum in Ontic Variables

Theorem 3.6 (FTD form of the sum)**.**

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{(n^2+1)^2} \;=\; \frac{2 + 4\ell \coth(4\ell) + 16\ell^2 \operatorname{csch}^2(4\ell)}{4}\,,} \tag{13}$$

where $\ell = \varpi^2/G^{*2}$.

Proof. Substitute $\pi = 4\ell$ into (11). □

Remark 3.7. *Every appearance of π in the standard result was a disguised occurrence of $4\varpi^2/G^{*2}$. The hyperbolic functions* coth *and* csch^2 *depend on a single transcendental argument: $4\ell = 4\varpi^2/G^{*2}$.*

3.3 Gelfond's Constant as the Exponential Scale

Proposition 3.8. *The exponential scale governing the hyperbolic functions in* (13) *is*

$$e^{4\ell} \;=\; e^{4\varpi^2/G^{*2}} \;=\; e^{\pi} \;\approx\; 23.1407\,,$$

which is Gelfond's constant, $(-1)^{-i}$.

Proof. $4\ell = \pi$ by Lemma 3.4; $e^{\pi} = (-1)^{-i}$ by Euler's identity. □

The nome of the Gaussian lattice $\mathbb{Z}[i]$ (with respect to the modular parameter $\tau = i$) is

$$q \;=\; e^{2\pi i\tau}\Big|_{\tau=i} \;=\; e^{-2\pi} \;=\; e^{-8\varpi^2/G^{*2}} \;\approx\; 1.87\times 10^{-3}\,.$$

The smallness of q (controlled entirely by ϖ^2/G^{*2}) is responsible for the rapid exponential suppression that makes the csch^2 correction contribute only $\sim$1.4% of the total sum, as we quantify in Section 6.

4 Connection to the Gaussian Lattice

4.1 The Sum as a Lattice Restriction

Observe that $n^2 + 1 = |n+i|^2$ for $n \in \mathbb{Z}$. Therefore:

$$S \;=\; \sum_{n=0}^{\infty} \frac{1}{|n+i|^4}\,. \tag{14}$$

The natural two-dimensional object associated to the Gaussian lattice $\mathbb{Z}[i]$ is the **Epstein zeta function**

$$Z_{\mathbb{Z}[i]}(s) \;=\; \sum_{z\in\mathbb{Z}[i]}{}' |z|^{-2s} \;=\; \sum_{(m,n)\in\mathbb{Z}^2}{}' (m^2+n^2)^{-s}\,, \tag{15}$$

where the prime denotes omission of $(m,n) = (0,0)$. Our sum S is the restriction of this lattice sum to the horizontal line $\mathrm{Im}(z) = 1$ (that is, m arbitrary with $n = 1$), further restricted to $m \geq 0$. Explicitly, the full $n = 1$ row of $Z_{\mathbb{Z}[i]}(2)$ contributes $\sum_{m=-\infty}^{\infty}(m^2+1)^{-2} = 2S - 1$, since the $m = 0$ term equals 1 and the remaining terms pair symmetrically.

Remark 4.1. *Unlike the unnormalized Eisenstein series* $G_4(\tau) = \sum'(m+n\tau)^{-4}$*, which involves the* fourth power *of a complex number, our sum involves the* squared modulus *raised to the second power:* $|m+ni|^{-4} = (m^2+n^2)^{-2}$*. These are distinct objects;* G_4 *is a holomorphic modular form, while the Epstein zeta function is real-valued and not holomorphic. The connection to modular forms passes through the identity* $Z_{\mathbb{Z}[i]}(s) = 4\,\zeta(s)\,L(s,\chi_{-4})$*, where* χ_{-4} *is the non-principal Dirichlet character modulo 4.*

4.2 The Eisenstein Series in Ontic Variables

Although our sum is a restriction of the Epstein zeta function rather than of G_4 directly, the Eisenstein series at the Gaussian point still controls the lattice's fundamental structure. The normalized Eisenstein series at $\tau = i$ is:

$$E_4(i) \;=\; 1 + 240\sum_{n=1}^{\infty}\sigma_3(n)\,q^n\,, \qquad q = e^{-2\pi}\,.$$

Theorem 4.2 (Eisenstein $E_4(i)$ in FTD form)**.**

$$E_4(i) \;=\; 3\Big(\frac{\varpi}{\pi}\Big)^4 \;=\; \frac{3\,G^{*8}}{256\,\varpi^4}\,. \tag{16}$$

Proof. The first equality is a classical result relating $E_4(i)$ to the lemniscate constant via the Weierstrass invariant g_2 of the square lattice (see Zucker [1]). For the second equality, substitute $\pi = 4\varpi^2/G^{*2}$:

$$3\left(\frac{\varpi}{\pi}\right)^4 = 3\left(\frac{\varpi\, G^{*2}}{4\varpi^2}\right)^4 = 3\left(\frac{G^{*2}}{4\varpi}\right)^4 = \frac{3\,G^{*8}}{256\,\varpi^4}\,. \qquad \square$$

Remark 4.3. *The full two-dimensional lattice structure at the Gaussian point is controlled entirely by ϖ and G^*. Our one-dimensional sum S is a restriction of this structure, and the restriction to a single row is what introduces the circular constant π as an intermediary through the bilateral-sum formula* (2)*.*

5 The Epstein–Lemniscatic Connection

5.1 The Epstein Factorisation in Ontic Variables

We now close the loop between the Epstein zeta function and the lemniscatic constants. The classical factorisation of $Z_{\mathbb{Z}[i]}(s)$ through Dirichlet series gives:

Theorem 5.1 (Epstein factorisation)**.**

$$Z_{\mathbb{Z}[i]}(s) \;=\; 4\,\zeta(s)\,L(s,\chi_{-4})\,, \tag{17}$$

where $\zeta(s)$ is the Riemann zeta function and $L(s,\chi_{-4})$ is the Dirichlet L-function attached to the non-principal character modulo 4 *(the Kronecker symbol $\chi_{-4}(n) = \left(\frac{-4}{n}\right)$).*

Proof. The number of representations $r_2(k) = \#\{(m,n) \in \mathbb{Z}^2 : m^2 + n^2 = k\}$ satisfies $r_2(k) = 4\sum_{d|k}\chi_{-4}(d)$, so

$$Z_{\mathbb{Z}[i]}(s) = \sum_{k=1}^{\infty} r_2(k)\,k^{-s} = 4\sum_{k=1}^{\infty}\Bigg(\sum_{d|k}\chi_{-4}(d)\Bigg)k^{-s} = 4\,\zeta(s)\,L(s,\chi_{-4})\,. \qquad \square$$

At $s = 2$, each factor acquires a specific identity:

Corollary 5.2 (Ontic form of $Z_{\mathbb{Z}[i]}(2)$)**.**

$$Z_{\mathbb{Z}[i]}(2) \;=\; 4\,\zeta(2)\,G \;=\; \frac{32\,\varpi^4}{3\,G^{*4}}\;\cdot\;G\,, \tag{18}$$

where $G = L(2,\chi_{-4}) \approx 0.915966$ is Catalan's constant.

Proof. Substitute $\zeta(2) = \pi^2/6$ and $\pi = 4\varpi^2/G^{*2}$:

$$\zeta(2) = \frac{\pi^2}{6} = \frac{16\varpi^4}{6\,G^{*4}} = \frac{8\varpi^4}{3\,G^{*4}}\,.$$

Then $Z_{\mathbb{Z}[i]}(2) = 4\cdot\dfrac{8\varpi^4}{3\,G^{*4}}\cdot G = \dfrac{32\,\varpi^4\,G}{3\,G^{*4}}$. $\square$

Remark 5.3 (The irreducible arithmetic residue)**.** *Catalan's constant*

$$G \;=\; \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} \;=\; 1 - \frac{1}{9} + \frac{1}{25} - \frac{1}{49} + \cdots \;\approx\; 0.915966$$

encodes the distribution of Gaussian primes: it measures the excess of primes $p \equiv 1 \pmod 4$ (which split in $\mathbb{Z}[i]$) over primes $p \equiv 3 \pmod 4$ (which remain inert), weighted by p^{-2}. This arithmetic content is intrinsic to the two-dimensional *Gaussian lattice and cannot be expressed in terms of ϖ and G^* alone. It is the irreducible arithmetic residue that distinguishes the norm projection $|z|^{-2s}$ from the holomorphic projection z^{-2s}.*

5.2 The Row Decomposition

The Epstein zeta function admits a natural decomposition by rows $n = \text{const}$:

$$Z_{\mathbb{Z}[i]}(2) \;=\; 2\,\zeta(4) + 2\sum_{n=1}^{\infty}\sum_{m=-\infty}^{\infty}\frac{1}{(m^2+n^2)^2}\,, \tag{19}$$

where $2\,\zeta(4) = \sum'_m m^{-4}$ is the $n = 0$ row and the factor of 2 in front of the sum accounts for the n and $-n$ rows.

Each bilateral row sum can be evaluated by differentiating (2) with respect to a and setting $a = n^2$:

$$\sum_{m=-\infty}^{\infty}\frac{1}{(m^2+n^2)^2} \;=\; \frac{\pi^2\operatorname{csch}^2(n\pi)}{2n^2} + \frac{\pi\coth(n\pi)}{2n^3}\,. \tag{20}$$

In ontic variables, each row depends on n through $\coth(4n\ell)$ and $\operatorname{csch}(4n\ell)$.

Proposition 5.4 (Row decomposition in ontic form)**.**

$$Z_{\mathbb{Z}[i]}(2) = \frac{2^8\,\varpi^8}{45\,G^{*8}} + 2\sum_{n=1}^{\infty}\left[\frac{8\varpi^4\operatorname{csch}^2(4n\ell)}{n^2\,G^{*4}} + \frac{2\varpi^2\coth(4n\ell)}{n^3\,G^{*2}}\right], \tag{21}$$

where $\ell = \varpi^2/G^{*2}$ and the leading term uses $\zeta(4) = \pi^4/90 = 2^8\varpi^8/(90\,G^{*8})$.

Remark 5.5 (Why the row restriction eliminates G)**.** *Each individual row sum (20) is expressible purely in terms of ϖ and G^* (via $\pi = 4\varpi^2/G^{*2}$). Catalan's constant enters only when we sum the row contributions and equate them to the Dirichlet-series factorisation $4\,\zeta(2)\,L(2,\chi_{-4})$. The row restriction to $n = 1$ that yields our sum S bypasses this global arithmetic content entirely: it uses only the Poisson summation formula over $\mathbb{Z}$, whose kernel is the Fourier transform of $1/(x^2+1)^2$ — a smooth function controlled by the single period ϖ.*

This explains why S admits a purely ontic closed form (Theorem 3.6), while the full $Z_{\mathbb{Z}[i]}(2)$ does not: the two-dimensional lattice sum encodes arithmetic information (the distribution of sums of two squares) that the one-dimensional restriction projects away.

5.3 Two Projections of One Lattice

The Gaussian lattice $\mathbb{Z}[i]$ supports two natural "projections" at weight 4, each revealing different content:

Projection	**Object**	**FTD form**	**Ontic purity**
Holomorphic (z^{-4})	$G_4(i),\ E_4(i)$	$3\,G^{*8}/(256\,\varpi^4)$	Fully ontic
Norm ($\lvert z\rvert^{-4}$)	$Z_{\mathbb{Z}[i]}(2)$	$32\,\varpi^4\,G/(3\,G^{*4})$	Ontic $\times$ arithmetic
Row restriction	S	$\dfrac{2+4\ell\coth 4\ell+16\ell^2\operatorname{csch}^2 4\ell}{4}$	Fully ontic

The holomorphic projection z^{-4} respects the four-fold symmetry of $\mathbb{Z}[i]$ and yields a holomorphic modular form whose value at $\tau = i$ is determined by the lattice periods ϖ and $i\varpi$ alone. The norm projection $|z|^{-4}$ discards phase information and introduces the arithmetic of $\mathbb{Z}[i]$ through the representation function $r_2(k)$, which carries Catalan's constant as its weight-2 residue.

Remark 5.6. *In both projections, the lemniscatic constants ϖ and G^* control the geometric structure of the lattice. The* arithmetic *factor G appears only when we ask a number-theoretic question ("how many lattice points lie on the circle $|z|^2 = k$?") rather than a geometric one ("what are the periods?"). The bridge constant G^* mediates the geometric content; Catalan's constant encodes the arithmetic residue.*

5.4 Numerical Verification

Quantity	Value	Method
S (direct summation, 10^5 terms)	1.306836975422908	Numerical
S (closed form, Eq. 11)	1.306836975422909	Analytic
S (FTD form, Eq. 13)	1.306836975422909	Analytic
π via $4\varpi^2/G^{*2}$	3.141592653589793	Verified
$E_4(i)$ (q-expansion, 500 terms)	1.455762892268709	Numerical
$3G^{*8}/(256\varpi^4)$	1.455762892268710	Analytic
$\zeta(2)$	1.644934066848226	Standard
$8\varpi^4/(3G^{*4})$	1.644934066848226	Ontic
Catalan $G = L(2, \chi_{-4})$	0.915965594177219	Standard
$Z_{\mathbb{Z}[i]}(2)$ via $4\,\zeta(2)\,G$	6.026812039691940	Factored
$32\varpi^4 G/(3G^{*4})$	6.026812039691940	Ontic
$Z_{\mathbb{Z}[i]}(2)$ via row decomposition	6.026812039691940	Row sum
$n=1$ bilateral row	1.613673950845818	Analytic
$2S - 1$	1.613673950845818	Verified

6 Structure of the Discrete–Continuous Bridge

The derivation passes through three distinct bridges, each moving from a more discrete to a more continuous description:

Step	From	To	Mechanism
1	Integer lattice $\mathbb{Z}$	Hyperbolic function of a	Bilateral sum / Poisson
2	Fixed sum at $a = 1$	Continuous family in a	Parameterisation
3	First-order sum	Squared denominator	Differentiation $-d/da$

The final answer decomposes into three additive contributions (within the numerator):

Term	Expression	Share	Interpretation
Constant	$2/4 = 0.500$	38.3%	Pure discrete residue
Linear in ℓ	$4\ell\coth(4\ell)/4$	60.3%	Bridge term (G^* mediation)
Quadratic in ℓ	$16\ell^2\operatorname{csch}^2(4\ell)/4$	1.4%	Curvature correction

The "constant" term $2/4 = 1/2$ is the part that would survive if the hyperbolic functions were replaced by their large-argument limits ($\coth \to 1$, $\operatorname{csch} \to 0$), corresponding to $\ell \to \infty$ or equivalently $q \to 0$. It represents the purely discrete contribution: the value the sum would approach if the lattice spacing were sent to zero relative to the period.

The "linear" term $4\ell\coth(4\ell)/4$ carries 60.3% of the answer and is dominated by the ratio ϖ^2/G^{*2}. This is the bridge term proper: it encodes the interaction between the discrete lattice and the continuous exponential.

The "quadratic" correction $16\ell^2\operatorname{csch}^2(4\ell)/4$ contributes only 1.4%. Its suppression is a direct consequence of the smallness of the nome $q = e^{-8\ell} \approx 1.87 \times 10^{-3}$, whose magnitude is set by $\ell = \varpi^2/G^{*2}$ alone.

7 The Genus-0 Projection

The bilateral sum formula (2) ultimately rests on the partial-fraction expansion of $\pi\cot(\pi z)$, which is a genus-0 (rational curve) identity. The hyperbolic functions coth and csch are all genus-0 objects: they parametrise the Riemann sphere $\mathbb{P}^1$.

The Gaussian lattice $\mathbb{Z}[i]$, however, is a genus-1 object. Its natural functions are the Weierstrass $\wp$-function and its derivatives, whose periods are ϖ and $i\varpi$, and whose invariants are expressible purely in terms of ϖ and G^*. (For the square lattice, $g_3 = 0$ by four-fold symmetry, and g_2 is determined by ϖ alone.)

The fact that the answer can be written in terms of coth and csch^2 of π is a consequence of the one-dimensional restriction collapsing the elliptic (genus-1) structure down to the circular (genus-0) one. Restricting to a single row $n = 1$ in the Epstein sum replaces the doubly-periodic lattice with a singly-periodic one, and single periodicity is the domain of trigonometric and hyperbolic functions.

In the FTD reading:

> π appears in this formula not because the sum is "about" circles, but because we chose to evaluate a lemniscatic object using circular tools. The ontic content is $\ell = \varpi^2/G^{*2}$. Everything else is projection.

8 The Dimensional Ladder

The genus-0 projection explains why π appears in the answer: we evaluated a two-dimensional (genus-1) object using one-dimensional (genus-0) tools. But this raises a deeper question: why π at all?

Euler's 1735 derivation of the Basel problem provides the answer. His method was to observe that $\sin(\pi z)/(\pi z)$ vanishes at every nonzero integer, and to treat it—by analogy with finite polynomials—as an infinite product over its roots:

$$\frac{\sin(\pi z)}{\pi z} = \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right). \tag{22}$$

Expanding the right side and comparing the z^2 coefficient with the Taylor series $\sin(\pi z)/(\pi z) = 1 - \pi^2 z^2/6 + \cdots$ gives $\sum 1/n^2 = \pi^2/6$.

Now substitute $\pi = 4\ell$, where $\ell = \varpi^2/G^{*2}$ is the ontic ratio:

$$\frac{\sin(4\ell z)}{4\ell z} = \prod_{n=1}^{\infty}\left(1 - \frac{z^2}{n^2}\right), \qquad \sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{(4\ell)^2}{6} = \frac{8\ell^2}{3} = \frac{8\varpi^4}{3G^{*4}}.$$

The right side of the product formula is unchanged—it depends only on the integers. The left side has replaced π with 4ℓ. The substitution is algebraically trivial, but it reveals the structural origin: the number $\pi^2/6$ is the zeta function of the one-dimensional integer lattice $\mathbb{Z}$ evaluated at $s = 2$. The zeros of $\sin(\pi z)$ are the lattice points of $\mathbb{Z}$. The Basel sum is a *lattice sum.*

This observation generalises. Each dimension has its own lattice, its own fundamental period, and its own zeta function:

D	Lattice	Period	Lattice zeta at $s = 2$	Product formula
1	$\mathbb{Z}$	$\pi = 4\ell$	$\zeta(2) = \dfrac{\pi^2}{6} = \dfrac{8\varpi^4}{3G^{*4}}$	$\dfrac{\sin \pi z}{\pi z} = \prod\left(1 - \dfrac{z^2}{n^2}\right)$
2	$\mathbb{Z}[i]$	ϖ	$Z_{\mathbb{Z}[i]}(2) = 4\,\zeta(2)\,L(2, \chi_{-4})$	Weierstrass $\sigma(z; \mathbb{Z}[i])$
3	$\mathbb{Z}^3$	G^*	$W_3 = \dfrac{G^{*2}}{2\pi} = \dfrac{\Gamma(1/4)^4}{4\pi^3}$	Watson integral

In $D = 1$, the "Euler move"—extending finite polynomial factorisations to infinite products—discovers π. The same move applied to the Gaussian lattice $\mathbb{Z}[i]$ in $D = 2$ discovers ϖ: the Weierstrass σ-function of $\mathbb{Z}[i]$ is the doubly-periodic analogue of $\sin(\pi z)$, and its periods are ϖ and $i\varpi$. In $D = 3$, the lattice Green's function of $\mathbb{Z}^3$—the Watson integral—discovers G^*, which absorbs the geometric scaling between the lemniscatic and circular domains.

Each lattice period is a "circle constant" for its dimension:

- π measures how far you walk around a circle (the boundary of the unit ball in $\mathbb{R}^1$ cross-section).
- ϖ measures how far you walk around a lemniscate (the fundamental domain boundary of $\mathbb{Z}[i]$).
- G^* measures the lattice Green's function of $\mathbb{Z}^3$—the propagator of the cubic lattice itself.

The reason π appears ubiquitously in mathematics is not that circles are fundamental. It is that $\mathbb{Z}$—the one-dimensional integer lattice—is the simplest lattice, and most classical formulas are computed over it. When we compute over $\mathbb{Z}[i]$, we find ϖ. When we compute over $\mathbb{Z}^3$, we find G^*.

The connection to FTD's dimension selection is immediate. The master quadratic $x^2 - 16{G^*}_D^2\, x + 16{G^*}_D^3 = 0$, applied to the Watson integral W_D in each dimension, gives $\lfloor x_-^{(D)} \rfloor = D$ only for $D = 3$ [3]. Among the infinite family of lattice periods (π, ϖ, G^*, ${G^*}_4$, $\ldots$), only the $D = 3$ member produces a self-consistent framework in which the number of color charges equals the spatial dimension.

The dimensional ladder is summarized in the following table:

D	Lattice	Period	Zeta function	$\lfloor x_-^{(D)} \rfloor \stackrel{?}{=} D$
1	$\mathbb{Z}$	π	$\zeta(2) = \pi^2/6$	N/A
2	$\mathbb{Z}[i]$	ϖ	$G_4(i) \propto \varpi^4$	**No** ($8 \neq 2$)
3	$\mathbb{Z}^3$	G^*	$W_3 = {G^*}^2/(2\pi)$	**Yes** ($3 = 3$)

Each dimension has its own "circle constant." π is the period of $\mathbb{Z}$. ϖ is the period of $\mathbb{Z}[i]$. G^* is the period of $\mathbb{Z}^3$. The relation $\pi = 4\varpi^2/{G^*}^2$ expresses the lowest-dimensional constant through the higher ones—π is the projection of lemniscatic geometry onto a line.

Euler's Basel problem is the one-dimensional shadow of a three-dimensional structure.

9 Summary of Rewriting

Quantity	**Standard Form**	**FTD (Ontic) Form**
Argument of coth / csch	π	$4\varpi^2/{G^*}^2$
Exponential scale	e^{π}	$e^{4\varpi^2/{G^*}^2}$
Lattice nome	$e^{-2\pi}$	$e^{-8\varpi^2/{G^*}^2}$
$E_4(i)$	$3(\varpi/\pi)^4$	$3{G^*}^8/(256\varpi^4)$
Prefactor π^2	π^2	$16\varpi^4/{G^*}^4$
$\zeta(2)$	$\pi^2/6$	$8\varpi^4/(3{G^*}^4)$
Euler product	$\sin(\pi z)/(\pi z) = \prod(1 - z^2/n^2)$	$\sin(4\ell z)/(4\ell z) = \prod(1 - z^2/n^2)$
$Z_{\mathbb{Z}[i]}(2)$	$4\,\zeta(2)\,G$	$32\varpi^4 G/(3{G^*}^4)$
Catalan G	$L(2, \chi_{-4})$	(irreducible arithmetic constant)

Conclusion. The discrete–continuous bridge in the evaluation of $\sum 1/(n^2+1)^2$ is real, but the conventional language in terms of π obscures its structural origin. The bridge is not π. The bridge is G^*.

The Epstein zeta factorisation (17) reveals an additional layer: the full two-dimensional lattice sum carries an irreducible arithmetic factor (Catalan's constant G) that encodes the distribution of Gaussian primes. This factor is eliminated by the row restriction that produces our one-dimensional sum S, explaining why S admits a purely ontic closed form while the full $Z_{\mathbb{Z}[i]}(2)$ does not.

Epistemic Classification

Following FTD conventions:

- **THEOREM** (exact algebraic): Equations (11), (13), (16), (17), (18), and Lemma 3.4. These are mathematically exact identities, verifiable by direct computation.
- **THEOREM** (standard number theory): The factorisation $Z_{\mathbb{Z}[i]}(2) = 4\,\zeta(2)\,L(2, \chi_{-4})$ and the identification of Catalan's constant as the irreducible arithmetic residue. The row-restriction elimination of the arithmetic factor follows from Poisson summation.
- **SELECTION PRINCIPLE** (pending proof): The claim that ϖ and G^* are ontologically prior to π. This is a well-motivated structural claim supported by the lemniscatic lattice framework but not provable within conventional mathematics, which treats all equivalent parametrisations as interchangeable.
- **HYPOTHESIS** (empirical fit): The identification of G^* as the mediator of all discrete–continuous transitions in physics. This is a broader FTD claim that this example illustrates but does not prove.

References

[1] I. J. Zucker, "The evaluation in terms of Γ-functions of the periods of elliptic curves admitting complex multiplication," *Math. Proc. Camb. Phil. Soc.* **82** (1977), 111–118.

[2] "The Ontic Derivation Chain: Twenty-Six Physical Constants from Two Inputs," companion paper (2026).

[3] "One Unit of Existence: The Lifecycle of a Hydrogen Atom from Lattice Geometry to Dissolution," companion paper (2026).

[4] "The Instantiating Formula: An Ontological Constructor for Foundational Ternary Dynamics," companion paper (2026).

[5] "The Lattice Engine: A Computational Implementation of Foundational Ternary Dynamics," companion paper (2026).